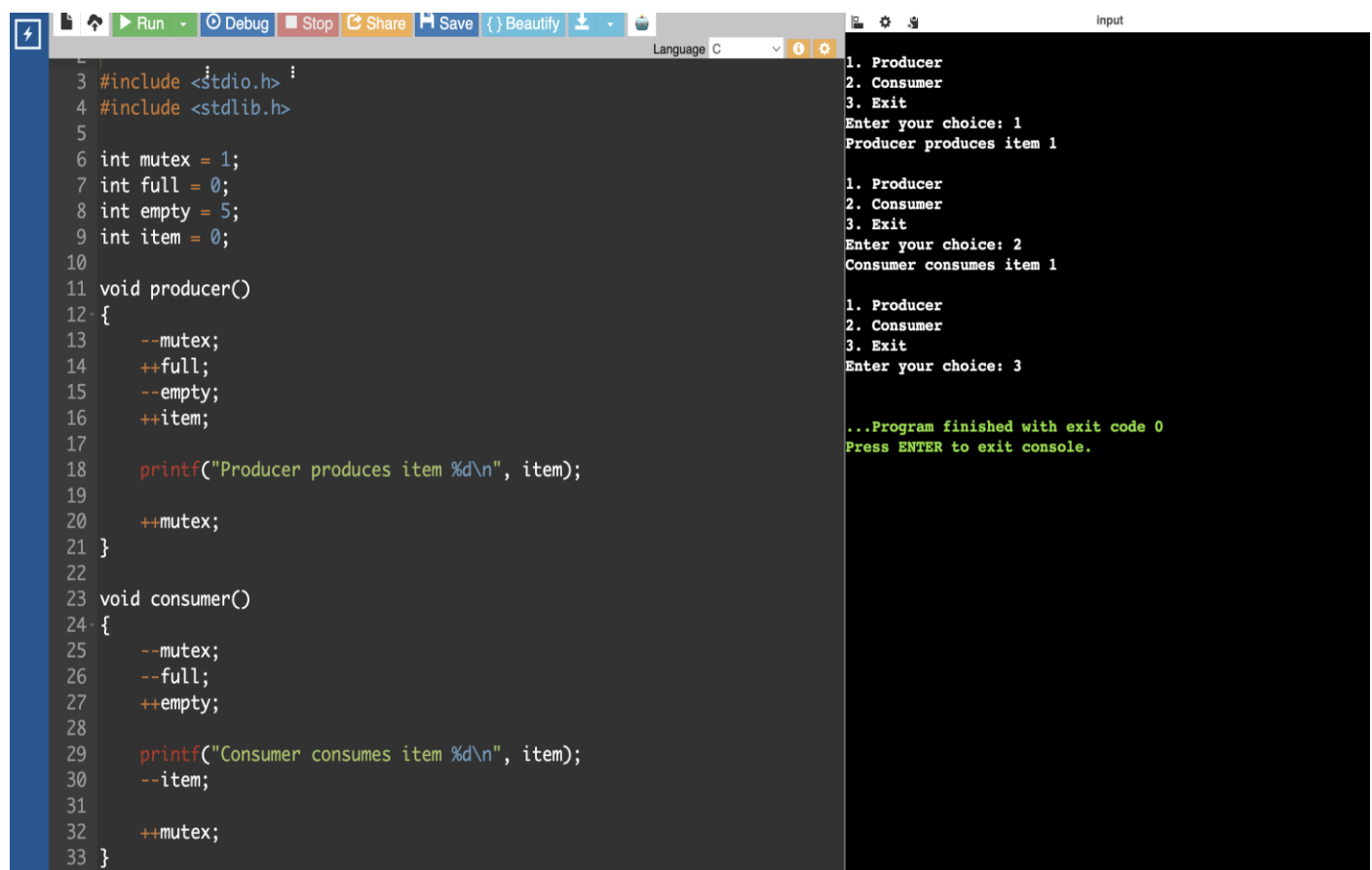


EXPERIMENT 29:

QUESTION:

Write a C program to simulate the solution of Classical Process Synchronization Problem

PROGRAM:



The image shows a code editor window with a C program for process synchronization. The program uses a mutex and shared counters to coordinate a producer and a consumer. The console output on the right shows the program's execution, including menu prompts, user choices, and item production/consumption messages.

```
1  #include <stdio.h>
2
3  #include <stdlib.h>
4
5
6  int mutex = 1;
7  int full = 0;
8  int empty = 5;
9  int item = 0;
10
11 void producer()
12 {
13     --mutex;
14     ++full;
15     --empty;
16     ++item;
17
18     printf("Producer produces item %d\n", item);
19
20     ++mutex;
21 }
22
23 void consumer()
24 {
25     --mutex;
26     --full;
27     ++empty;
28
29     printf("Consumer consumes item %d\n", item);
30     --item;
31
32     ++mutex;
33 }
```

1. Producer
2. Consumer
3. Exit
Enter your choice: 1
Producer produces item 1
1. Producer
2. Consumer
3. Exit
Enter your choice: 2
Consumer consumes item 1
1. Producer
2. Consumer
3. Exit
Enter your choice: 3
...Program finished with exit code 0
Press ENTER to exit console.